

How to estimate the supply-price elasticity if no measure for the housing stock Q is available?

Proposal by Malpezzi & Macleannan (2001, Journal of Housing Economics, No. 10, 278-306) (slightly adjusted)

Consider the following three equation model (this is very similar to the lecture example):

$$Q_D = \alpha_0 + \alpha_1 P_h + \alpha_2 Y + \alpha_3 D + e$$

$$Q_s = \beta_0 + \beta_1 P_h + \mu$$

$$Q_D = Q_s$$

Y is the income, D is a vector of additional demand shifters.

If we set $Q_D = Q_s$, then P_h reduced form can be written as:

$$P_h = \frac{\alpha_0 - \beta_0}{\beta_1 - \alpha_1} + \frac{\alpha_2}{\beta_1 - \alpha_1} Y + \frac{\alpha_3}{\beta_1 - \alpha_1} D + \pi$$

If we set: $\gamma_0 = \frac{\alpha_0 - \beta_0}{\beta_1 - \alpha_1}$; $\gamma_1 = \frac{\alpha_2}{\beta_1 - \alpha_1}$; $\gamma_2 = \frac{\alpha_3}{\beta_1 - \alpha_1}$

Then we can write:

$$P_h = \gamma_0 + \gamma_1 Y + \gamma_2 D + \pi$$

What we can estimate easily is γ_0 , γ_1 and γ_2

Say we estimate a $\gamma_1 = 0.375$

What we are interested in, however, is β_1 (the supply price elasticity).

Unfortunately, we can't estimate the three equation model above using TSLS if we don't have any information on Q.

However, there are estimates in the literature on the:

- Price elasticity of housing demand (α_1) and
- Long-run income elasticity of demand (α_2)

The former value typically is b/w -0.5 and -1 and the latter value typically is b/w 0.5 and 1.

Say if we assume that $\alpha_1 = -0.75$ and $\alpha_2 = 0.75$ then, we can back out

$$\beta_1 = \frac{\alpha_2}{\gamma_1} + \alpha_1$$

Using the values above

$$\begin{aligned}\beta_1 &= \frac{0.75}{0.375} - 0.75 \\ &= 1.25\end{aligned}$$

(Since we know γ_1 from estimating the reduced form specification, we can derive the supply price elasticity β_1 .)

Obviously, this approach relies on α_1 and α_2 to be rather accurate.

One can do robustness checks to see how sensitive the estimated supply-price elasticity is to changes in the assumption on α_1 and α_2 .

If we assume the min value for α_1 (-0.5) and the max value for α_2 (1) we would get a

$$\begin{aligned}\beta_1 &= \frac{1}{0.375} - 0.5 \\ &= 2.167\end{aligned}$$

If we assume the max value for α_1 (-1) and the min value for α_2 (0.5) we would get a

$$\begin{aligned}\beta_1 &= \frac{0.5}{0.375} - 1 \\ &= 0.333\end{aligned}$$

Obviously this is a very wide range (not very comforting)